



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

**Dept. of Computer Science
Faculty of Science and Technology**

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Spring 2024-2025

Section: [O]

Group No: 08

Project Report On

Travel Agency Management System [TripVisor]

Supervised By

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Submitted By:

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CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				Total =
Requirement fulfillment	Properly demonstrate a real-life scenario-based project with proper functional requirement identification for the Object-Oriented Programming project development activities.				
Validation	Ensuring the ability of students' proper demonstration on validation forms in their system in terms of dealing with the data.				
Verification	Identifying if the students can verify the system data along with proper functional requirements in terms of data flow.				

INTRODUCTION

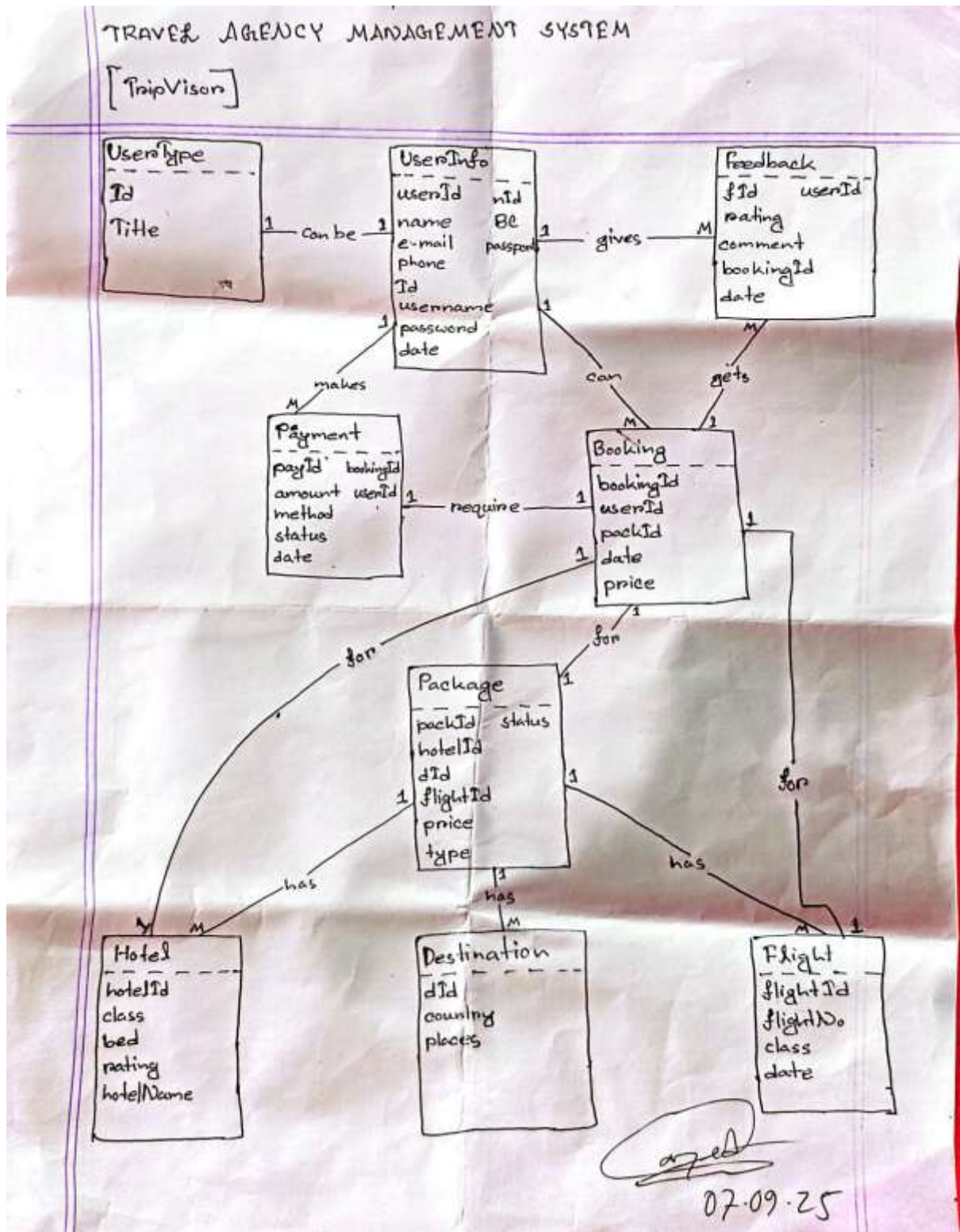
This project is a Windows Forms (.NET Framework) application in C# called Travel Agency Management System. It is designed to make travel planning easier for users and help agencies manage bookings more efficiently.

In this system, users can explore different travel packages and choose their desired destinations. They can also book hotels directly through the system without visiting multiple websites or contacting agents. The application stores all information such as package details, customer bookings, and hotel data in one place, making it simple and organized.

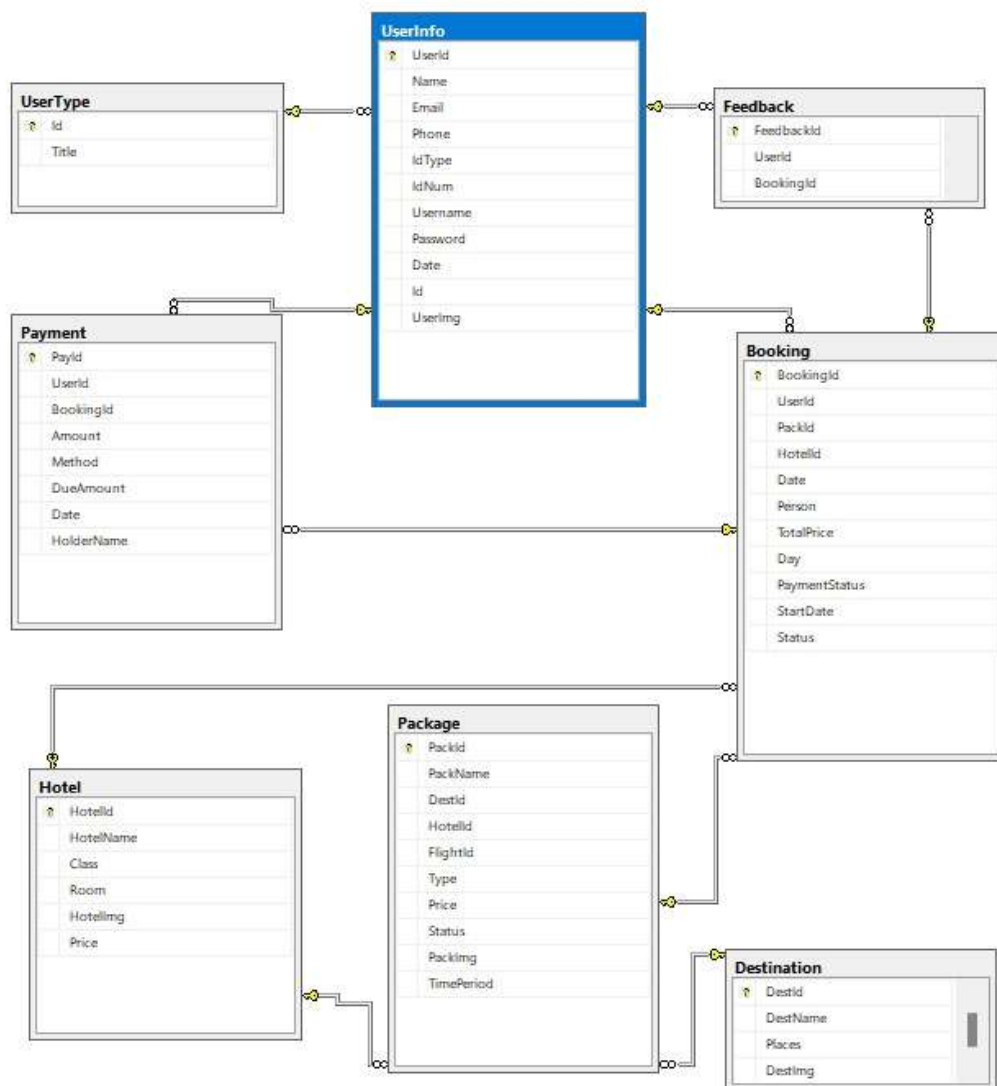
For the admin side, the system allows managing packages, destinations, hotel lists, and customer information easily. It helps reduce manual work and errors while saving time for both customers and travel managers.

Overall, this project provides a user-friendly way to plan trips, select travel packages, and book hotels — all in one system.

SCHEMA DIAGRAM



Final



FINALIZATION TABLE

1) UserType

Id(PK), Title

2) UserInfo

Id(PK), UserId, Phone, Email, IdType, IdNum, Username, Password, Date, Id(FK)

3) Destination

DestId(PK), DestName, Places, Destination

4) Hotel

HotelId(PK), HotelName, Class, Room, HotelImg, Price

5) Package

PackId(PK), PackName, DestId(FK), HotelId(FK), FlightId(FK), Type, Price, Status, PackImg, TimeAndDate

6) Booking

BookingId(PK), UserId(FK), PackId(FK), HotelId(FK), Date, Person, TotalPrice, Day, PaymentStatus, StartDate

7) Payment

PayId(PK), UserId(FK), BookingId(FK), Amount, Method, DueAmount, Date, HolderName

TABLE FROM DATABASE

Table : UserInfo

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the database structure, including the 'UserInfo' table. The right pane shows the 'Current connection parameters' for the 'LAPTOP-LSBMDQ-128' server. The main pane displays the 'UserInfo' table structure and data.

Table Structure:

```

CREATE TABLE UserInfo (
    ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Email VARCHAR(100),
    Phone VARCHAR(20),
    Address VARCHAR(200),
    City VARCHAR(50),
    State VARCHAR(50),
    Zip VARCHAR(10),
    Password VARCHAR(50),
    DateOfBirth DATE,
    Gender VARCHAR(10),
    MaritalStatus VARCHAR(10),
    Religion VARCHAR(50),
    Education VARCHAR(50),
    Occupation VARCHAR(50),
    LastLogin DATETIME,
    IsActive BIT,
    IsDeleted BIT
)

```

Data:

ID	Name	Email	Phone	Address	City	State	Zip	Password	DateOfBirth	Gender	MaritalStatus	Religion	Education	Occupation	LastLogin	IsActive	IsDeleted
1	John Doe	john.doe@gmail.com	9178805110	123 Main St	NY	10001	10001	12345678	1990-01-01	M	Married	Catholic	Bachelor's	Software Engineer	2023-04-10 10:00:00	1	0
2	Jane Smith	jane.smith@gmail.com	9178805111	456 Elm St	CA	90001	90001	87654321	1985-05-15	F	Single	Protestant	Master's	Marketing Manager	2023-03-20 09:00:00	1	0
3	Michael Brown	michael.brown@gmail.com	9178805112	789 Oak St	TX	75001	75001	11223344	1992-08-22	M	Married	Muslim	High School	Teacher	2023-04-05 11:00:00	1	0
4	Sarah Lee	sarah.lee@gmail.com	9178805113	101 Pine St	FL	33101	33101	55667788	1988-11-10	F	Single	Buddhist	PhD	Research Scientist	2023-04-08 14:00:00	1	0
5	David Kim	david.kim@gmail.com	9178805114	202 Birch St	WA	98101	98101	99887766	1995-03-05	M	Married	Korean	College	Student	2023-04-09 16:00:00	1	0
6	Emily White	emily.white@gmail.com	9178805115	303 Cedar St	IL	60601	60601	44332211	1991-07-18	F	Single	Jewish	Master's	Data Analyst	2023-04-10 08:00:00	1	0
7	Robert Green	robert.green@gmail.com	9178805116	404 Maple St	OH	43201	43201	33221100	1987-09-03	M	Married	Catholic	Bachelor's	Sales Representative	2023-04-07 12:00:00	1	0
8	Lisa Black	lisa.black@gmail.com	9178805117	505 Elm St	MI	48101	48101	22110099	1993-12-25	F	Single	Anglican	High School	Retail Worker	2023-04-10 10:00:00	1	0
9	James Taylor	james.taylor@gmail.com	9178805118	606 Oak St	PA	19101	19101	11009988	1989-06-12	M	Married	Quaker	Master's	Professor	2023-04-09 15:00:00	1	0
10	Alice Johnson	alice.johnson@gmail.com	9178805119	707 Pine St	NC	27601	27601	00998877	1994-02-28	F	Single	Methodist	Bachelor's	Librarian	2023-04-10 09:00:00	1	0
11	Kevin Davis	kevin.davis@gmail.com	9178805120	808 Cedar St	MO	64101	64101	99887766	1990-10-10	M	Married	Baptist	High School	Truck Driver	2023-04-08 13:00:00	1	0
12	Nancy Wilson	nancy.wilson@gmail.com	9178805121	909 Elm St	NE	68101	68101	88776655	1986-04-04	F	Single	Episcopal	College	Writer	2023-04-10 11:00:00	1	0
13	Christopher Moore	christopher.moore@gmail.com	9178805122	1010 Oak St	DE	19701	19701	77665544	1997-01-01	M	Single	Presbyterian	High School	IT Support	2023-04-09 10:00:00	1	0
14	Michelle Carter	michelle.carter@gmail.com	9178805123	1111 Pine St	SD	57101	57101	66554433	1992-05-15	F	Single	Lutheran	Bachelor's	Graphic Designer	2023-04-10 12:00:00	1	0
15	Daniel Evans	daniel.evans@gmail.com	9178805124	1212 Cedar St	WY	82101	82101	55443322	1988-09-20	M	Married	Evangelical	High School	Construction Worker	2023-04-07 14:00:00	1	0
16	Olivia Baker	olivia.baker@gmail.com	9178805125	1313 Elm St	VT	05401	05401	44332211	1995-11-03	F	Single	Unitarian	College	Freelance Writer	2023-04-10 07:00:00	1	0
17	Benjamin Hall	benjamin.hall@gmail.com	9178805126	1414 Oak St	RI	02901	02901	33221100	1991-03-18	M	Married	Anglican	Bachelor's	Software Developer	2023-04-09 16:00:00	1	0
18	Sophia King	sophia.king@gmail.com	9178805127	1515 Pine St	CT	06101	06101	22110099	1993-07-22	F	Single	Methodist	High School	Event Planner	2023-04-10 13:00:00	1	0
19	Matthew Scott	matthew.scott@gmail.com	9178805128	1616 Cedar St	MD	21201	21201	11009988	1989-12-01	M	Married	Catholic	Bachelor's	Business Analyst	2023-04-08 11:00:00	1	0
20	Isabella Adams	isabella.adams@gmail.com	9178805129	1717 Elm St	MA	02101	02101	00998877	1996-06-10	F	Single	Presbyterian	College	Marketing Specialist	2023-04-10 14:00:00	1	0
21	Julian Taylor	julian.taylor@gmail.com	9178805130	1818 Oak St	NY	10001	10001	99887766	1994-02-28	M	Single	Anglican	Bachelor's	Software Engineer	2023-04-09 15:00:00	1	0

Table : UserType

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the database structure, including the 'UserType' table. The right pane shows the 'Current connection parameters' for the 'LAPTOP-LSBMDQ-128' server. The main pane displays the 'UserType' table structure and data.

Table Structure:

```

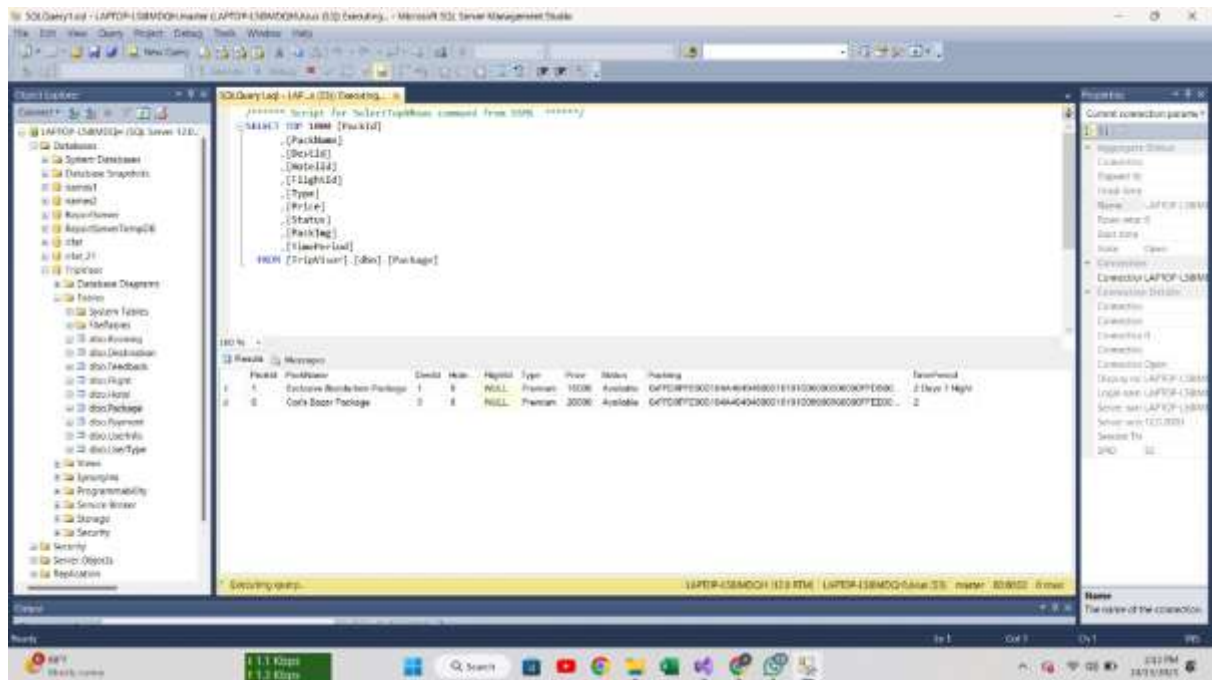
CREATE TABLE UserType (
    ID INT PRIMARY KEY,
    Name VARCHAR(50)
)

```

Data:

ID	Name
1	Admin
2	Manager
3	Employee
4	Customer

Table : Package



SQL Query Log - LAPTOP-LSBMOGH (SQL Server 12.0) Executing...

```

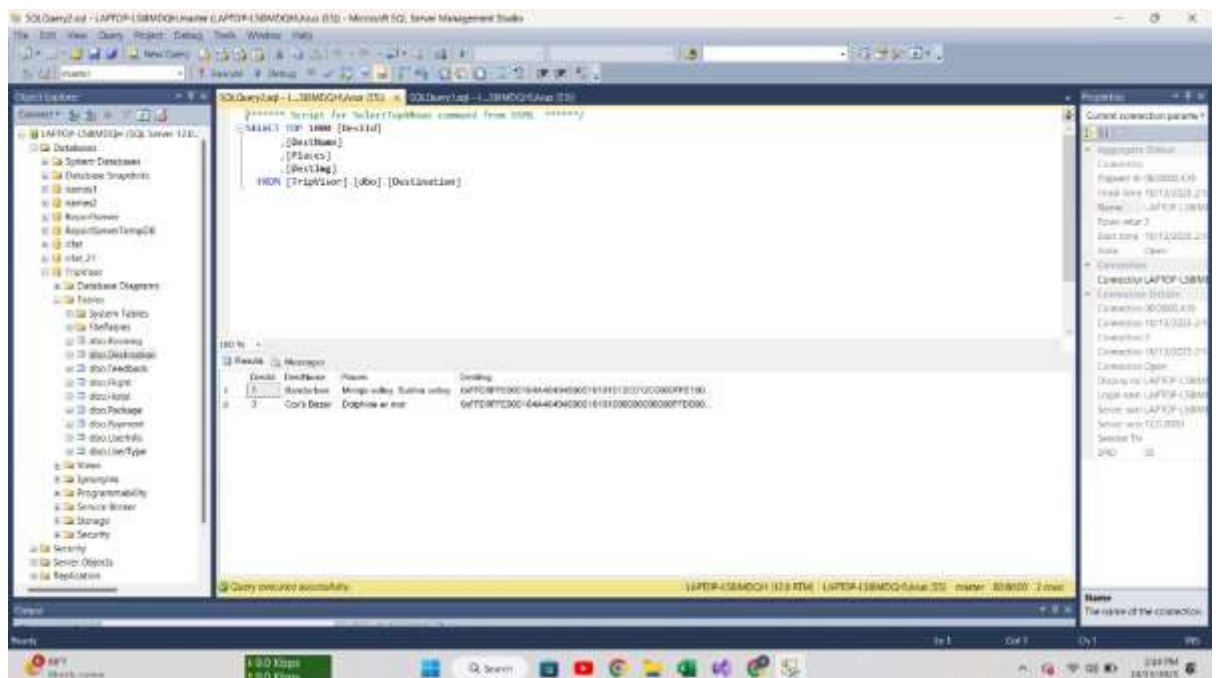
--===== Script for SelectTopPackage command from SQLS *****
SELECT TOP 1000 [PackageID]
,[PackageName]
,[Description]
,[Price]
,[Status]
,[PackageType]
FROM [TripVision].[dbo].[Package]

```

PackageID	PackageName	Description	Price	Status	PackageType
1	Exclusive Romantic Package	1	0	NULL	Premium
2	Cash Back Package	2	0	NULL	Premium

Query executed successfully.

Table : Destination



SQL Query Log - LAPTOP-LSBMOGH (SQL Server 12.0) Executing...

```

--===== Script for SelectTopDestination command from SQLS *****
SELECT TOP 1000 [DestinationID]
,[DestinationName]
,[DestinationType]
FROM [TripVision].[dbo].[Destination]

```

DestinationID	DestinationName	DestinationType
1	Banarash	Mango valley
2	Cash Back	Daphne in ear

Query executed successfully.

Table : Hotel

SQL Server Enterprise Manager - LAPTOP-LSBMOGNAME (LAPTOP-LSBMOGNAME) (RT) - Microsoft SQL Server Management Studio

Query: SELECT * FROM [Hotel] (HotelID)

HotelID	HotelName	Class	Room	Rate	Price
1	Hotel 1	High	Super Deluxe	1000	1000
2	Hotel 2	Low	Double Bed	500	500
3	Hotel 3	High	Single	1500	1500

Table : Booking

SQL Server Enterprise Manager - LAPTOP-LSBMOGNAME (LAPTOP-LSBMOGNAME) (RT) - Microsoft SQL Server Management Studio

Query: SELECT * FROM [Booking] (BookingID)

BookingID	HotelID	Room	Rate	Price	Status
1	1	1	1000	1000	Confirmed
2	1	2	500	500	Confirmed
3	1	3	1500	1500	Confirmed
4	2	1	500	500	Confirmed
5	2	2	500	500	Confirmed
6	2	3	500	500	Confirmed
7	3	1	1500	1500	Confirmed
8	3	2	1500	1500	Confirmed
9	3	3	1500	1500	Confirmed
10	1	1	1000	1000	Pending
11	1	2	500	500	Pending
12	1	3	1500	1500	Pending
13	2	1	500	500	Pending
14	2	2	500	500	Pending
15	2	3	500	500	Pending
16	3	1	1500	1500	Pending
17	3	2	1500	1500	Pending
18	3	3	1500	1500	Pending

Table : Payment

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the database structure, including the 'Payment' table. The right pane shows the 'Payment' table structure with the following columns: PaymentId, BookingId, Amount, Method, Location, Date, and PaymentType. The bottom pane displays the query result for the 'Payment' table, showing 4 rows of data.

PaymentId	BookingId	Amount	Method	Location	Date	PaymentType
1	10	20	Credit	10	2023-10-10 11:22:28.773	gpt
2	11	20	Credit	10	2023-10-10 11:22:28.773	gpt
3	12	20	Credit	10	2023-10-10 11:22:28.773	gpt
4	13	20	Credit	10	2023-10-10 11:22:28.773	gpt

Table : Feedback

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the database structure, including the 'Feedback' table. The right pane shows the 'Feedback' table structure with the following columns: FeedbackId, BookingId, Comment, and FeedbackType. The bottom pane displays the query result for the 'Feedback' table, showing 4 rows of data.

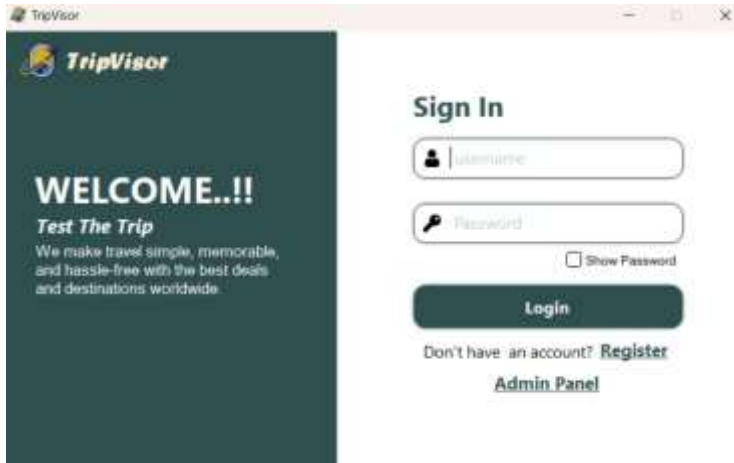
FeedbackId	BookingId	Comment	FeedbackType
1	10	Good	1
2	11	Good	1
3	12	Good	1
4	13	Good	1

FEATURES

Here are some features of our project “Travel Agency Management System”

- The system allows the Owner, Manager, Employee, and Customer to log in with secure credentials, each having different levels of access and control.
- The Owner has full authority to manage all system operations, including packages, bookings, employees, customers, and financial transactions.
- The Manager can handle travel packages, hotel partnerships, customer bookings, and oversee employees’ activities.
- The Employee can assist in managing bookings, handling customer queries, and updating package or hotel details as instructed by the manager or owner.
- The Customer can browse available travel packages, view destination details, book desired packages, make payments, and cancel bookings if necessary.
- The system supports real-time package booking, allowing customers to select destinations, hotel options, and travel dates based on availability.
- A secure login system ensures that data and operations are protected, restricting unauthorized access to sensitive information.
- The payment module records transaction details such as amount, payment method, and date, ensuring transparent and accurate records.
- The system maintains a hotel management section, where hotel details, room availability, and price lists can be updated by the admin or manager.
- The owner and manager can generate reports summarizing bookings, payments, and customer activity for analysis and decision-making.
- The system provides notifications and updates about booking confirmations, cancellations, or payment status to customers.
- With a Windows Forms (.NET Framework) interface, the system ensures smooth navigation, ease of use, and a friendly user experience.
- The database stores all information—packages, bookings, customers, employees, and hotels—ensuring accuracy, integrity, and consistency across the system.
- The system helps reduce manual paperwork and errors by automating travel planning, hotel booking, and payment processes.
- All records and data are dynamically retrieved from the database to ensure that information displayed in the application is always up to date.

Project Screenshot (User wise)



The screenshot shows the TripVisor login interface. On the left, a dark green sidebar contains the TripVisor logo and a 'WELCOME...!!' message with the tagline 'Test The Trip' and a brief description of the service. The main area is white and features a 'Sign In' section with input fields for 'Username' and 'Password'. A 'Show Password' checkbox is located below the password field. A dark green 'Login' button is positioned below the inputs. At the bottom, there are links for 'Don't have an account? Register' and 'Admin Panel'.

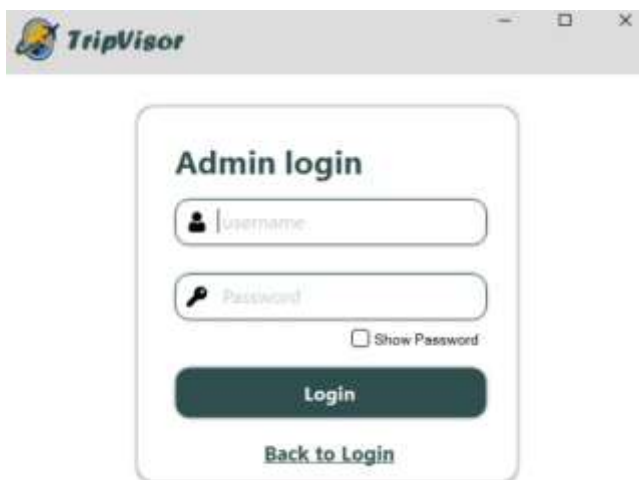
Login Page



The screenshot displays the TripVisor sign-up page. It includes a 'Sign UP' header and input fields for 'Full Name', 'Email', and 'Phone'. Below these is a 'Select ID Type:' section with radio buttons for 'NO', 'Both Certificate', and 'Passport'. Further down are fields for 'Submitted ID No.', 'Username', and 'Password'. A dark green 'Register' button is at the bottom, with a 'Back to Login' link underneath.

Resister Page

Admin Page



The screenshot shows the TripVisor admin login page. It features a dark green sidebar with the TripVisor logo and navigation links for 'User Info', 'Manager', and 'All bookings'. The main area is white and contains an 'Admin login' section with 'Username' and 'Password' input fields, a 'Show Password' checkbox, and a dark green 'Login' button. A 'Back to Login' link is at the bottom.

Admin Login



The screenshot displays the TripVisor admin panel for an owner. It features a dark green sidebar with navigation links for 'User Info', 'Manager', 'All bookings', and 'Logout'. The main area is white and contains a table with columns for 'Date', 'Manager', 'ShipType', 'Customer', 'Status', 'Price', 'Kilop', 'Miles', 'Status', 'Status', 'Date', 'Time', and 'Notes'. The table is currently empty, and a 'Refresh' button is located at the bottom right.

Date	Manager	ShipType	Customer	Status	Price	Kilop	Miles	Status	Status	Date	Time	Notes
------	---------	----------	----------	--------	-------	-------	-------	--------	--------	------	------	-------

Admin Panel [Owner]



Admin Panel [Manager]



Admin Panel [Employee]



Manage Package



Manage Destination



Manage Hotel

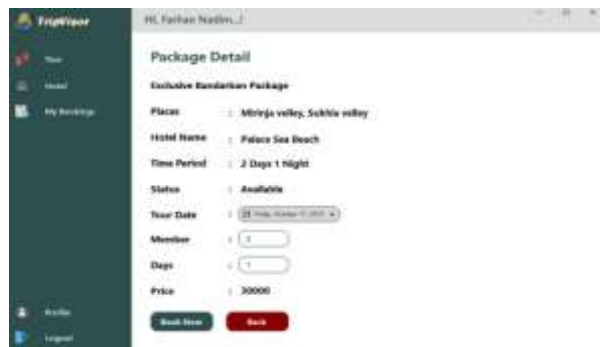


Manage Bookings

Customer View



Package Page



Package Details



Hotel Page



Hotel Details



Payment Page



Customer Booking Page

CONCLUSION:

The Travel Agency Management System project was successfully developed and implemented to provide a centralized, efficient platform for managing travel packages, customer bookings, and financial transactions.

The system's key functionalities—including the display of diverse travel packages, secure online booking, integrated payment processing, and a dedicated feedback mechanism—directly address the agency's needs for streamlined operations and enhanced customer service. By replacing manual processes with an automated system, we have reduced data redundancy, minimized human error, and ensured consistent record-keeping.

In conclusion, this project effectively leverages database management and web technologies to deliver a robust and scalable solution that significantly modernizes the travel agency's workflow, leading to improved efficiency and a better customer experience.

REFERENCES

- A. Troelsen and P. Japikse, Pro C# 10 with .NET 6: Foundational Principles and Practices in Programming, Apress, 2022.
- S. Rana and R. Chatterjee, "Design and Implementation of a Travel Agency Management System using Database-Oriented Approach," International Journal of Computer Applications, vol. 183, no. 15, pp. 25–30, 2021. [Online]. Available: <https://doi.org/10.5120/ijca2021921541>
- Microsoft, Windows Forms Overview, Microsoft Learn, 2025. [Online]. Available: <https://learn.microsoft.com/en-us/dotnet/desktop/winforms/>
- Microsoft, ADO.NET Overview, Microsoft Learn, 2025. [Online]. Available: <https://learn.microsoft.com/en-us/dotnet/framework/data/adonet/ado-net-overview>
- GUNAUI2 Framework
- A. Al-Mamun and M. Rahman, "Design and Development of a Travel Agency Management System," International Journal of Computer Science and Mobile Computing, vol. 9, no. 5, pp. 12–18, 2020.
- R. Lafore, Object-Oriented Programming in C++, Pearson, 2018.
- I. Sommerville, Software Engineering, 10th ed., Pearson, 2016.

